# Olympiad Geometry – III

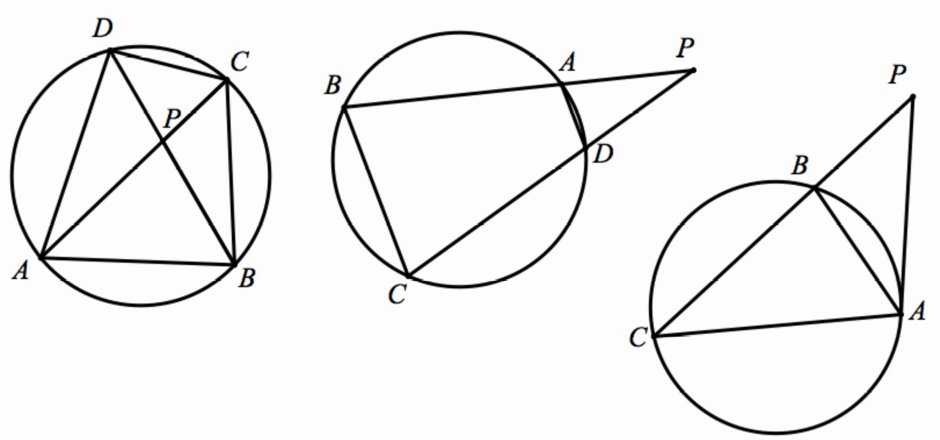
## 5. Power of a point to a circle

### Intersecting chord theorem

(a) Let $ABCD$ be a convex quadrilateral and let $P = AC \cap BD$. Then the points $A$, $B$, $C$, $D$ are concyclic if and only if $PC \cdot PA = PB \cdot PD$.

(b) Let $ABCD$ be a convex quadrilateral and let $P = AB \cap CD$. Then the points $A$, $B$, $C$, $D$ are concyclic if and only if $PA \cdot PB = PC \cdot PD$.

(c) Assume points $P$, $B$, $C$ are collinear in this order and point $A$ does not lie on this line. Then the line $PA$ is tangent to the circumcircle of triangle $ABC$ if and only if $PA^2 = PB \cdot PC$.



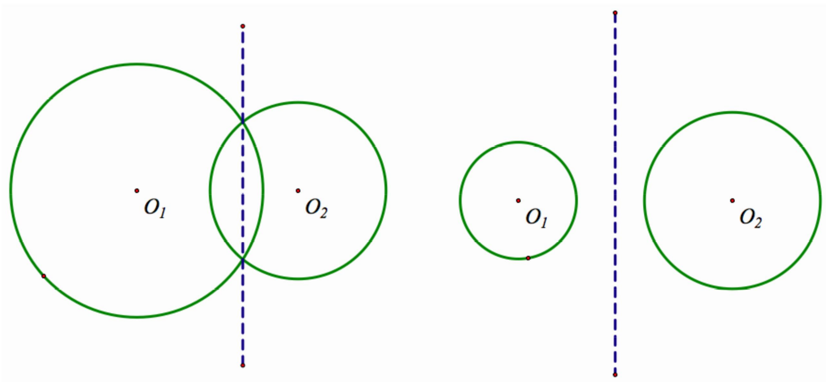

### Power of a Point

Given point $P$ and circle $\omega$, let be an arbitrary line passing through $P$ and intersecting $\omega$ at points $A$ and $B$. Then the value of $PA \times PB$ does not depend on the choice of . Also, if $P$ lies outside of $\omega$ and $PT$, $T \in \omega$, is a tangent to $\omega$ then $PA \times PB = PT^2$.

If we denote the centre of $\omega$ by $O$ and its radius by $R$ then $PA \times PB = |\, OP^2 - R^2 \,|$.

The quantity $p(P, \omega) = OP^2 - R^2$ is called the power of point $P$ with respect to circle $\omega$.

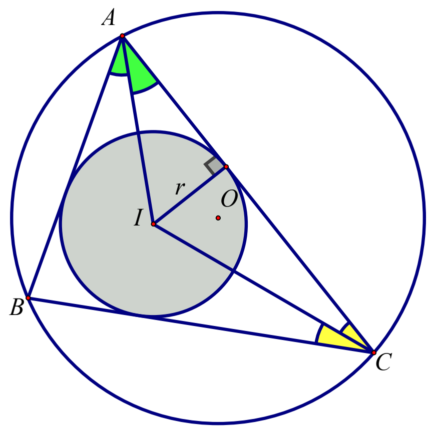
Given two circles $\omega_1$ and $\omega_2$, the locus of points for which they have the same power of point with respect to the circles is a line called the radical axis of the two circles




In a rectangle coordinate system, the power of a point $P(a, b)$ to the circle $x^2 + y^2 + Dx + Ey + F = 0$ is given by $|\, a^2 + b^2 + Da + Eb + F \,|$.

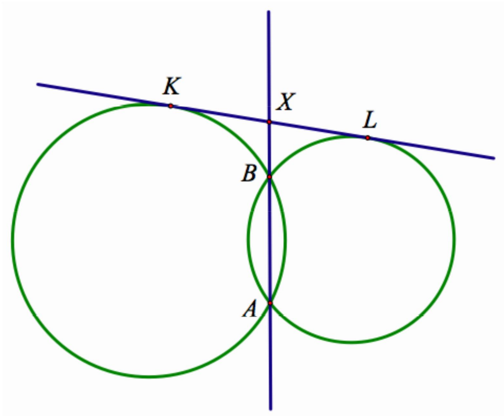
**Euler's Identity**

In $\Delta ABC$ with circumcentre $O$ and incentre $I$, we have $OI^2 = R^2 - 2Rr$ and hence $R \geq 2r$ equality holds if and only if $ABC$ is equilateral.



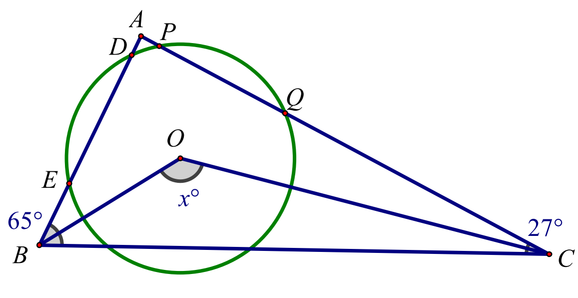

**Example 5.1 (Power of a point)**

Let the circles $\omega_1$, $\omega_2$intersect at points $A$, $B$.
Denoted by $K$, $L$ the points of tangency of the common external tangent with circles $\omega_1$, $\omega_2$ respectively. Then the line $AB$ bisects the segment $KL$.

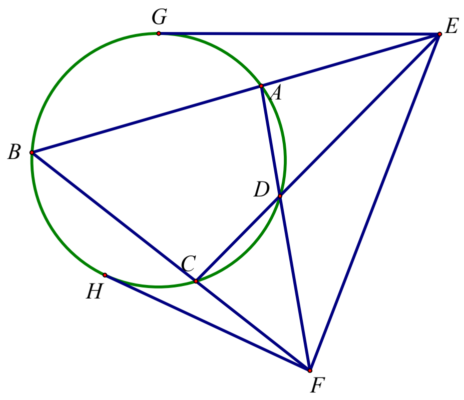



**Exercise:**

1. **(2019 PCIMC Heat Event (Senior) #20)**
   In $\Delta ABC$, $\angle ABC = 65^\circ$ and $\angle ACB = 27^\circ$. A circle with centre $O$ intersects $AB$ at $D$ and $E$ and intersects $AC$ at $P$ and $Q$. If $AD \times AE = BD \times BE$, $AP \times AQ = CP \times CQ$ and $\angle BOC = x^\circ$, find the value of $x$.

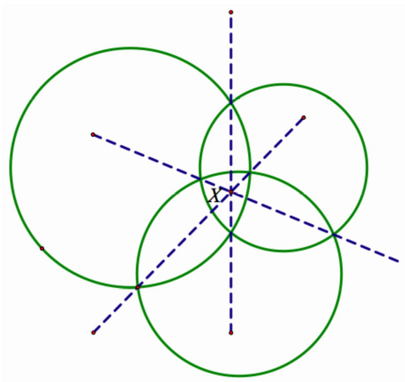


2. In the figure, $EG$ and $FH$ are tangents to the circle. Prove that $EF^2 = EG^2 + FH^2$



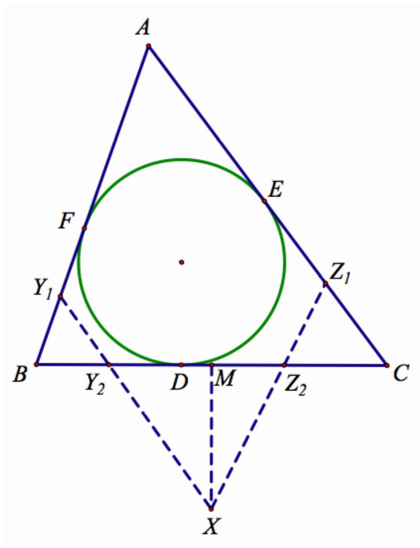

**Radical Centre**

Let $\omega_1$, $\omega_2$, $\omega_3$ be circles with pairwise distinct centres. Then their pairwise radical axes are either parallel or concurrent. The point of occurrence is called the radical centre of the three circles.



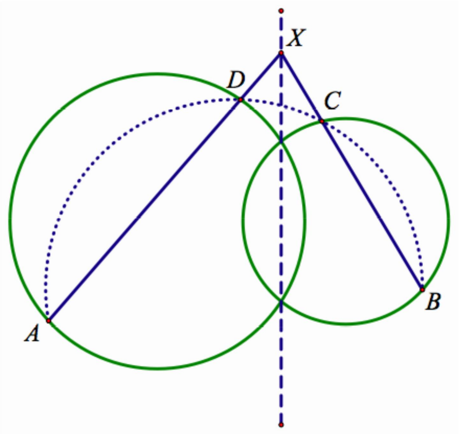

**Example 5.2 (Radical Centre)**

Let the incircle $\omega$ of triangle $ABC$ touch $BC$, $CA$ and $AB$ at $D$, $E$ and $F$ respectively. Let $Y_1$, $Y_2$, $Z_1$, $Z_2$ and $M$ be the midpoints of $BF$, $BD$, $CE$, $CD$ and BC respectively. Let $Y_1Y_2 \cap Z_1Z_2 = X$. Prove that $MX \perp BC$.



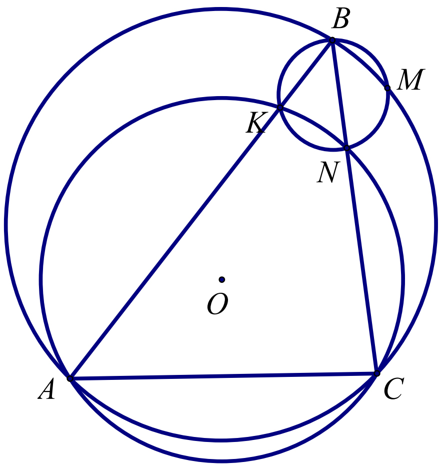
**Radical Lemma****

Let the line $L$ be radical axis of the circles $\omega_1$, $\omega_2$. Let $A$, $D$ be distinct points on $\omega_1$ and let $B$, $C$ be distinct points on $\omega_2$such that the lines $AD$ and $BC$ are not parallel. Then the lines $AD$ and $BC$ intersect at $L$ if and only if $ABCD$ is cyclic.




**Example 5.3 (IMO 1985 P5)**

A circle with centre $O$ passes through the vertices $A$ and $C$ of triangle $ABC$ and intersects the segments $AB$ and $BC$ again at distinct points $K$ and $N$, respectively. The circumscribed circles of the triangles $ABC$ and $KBN$ intersect at exactly two distinct points $B$ and $M$. Prove that angle $OMB$ is a right angle.



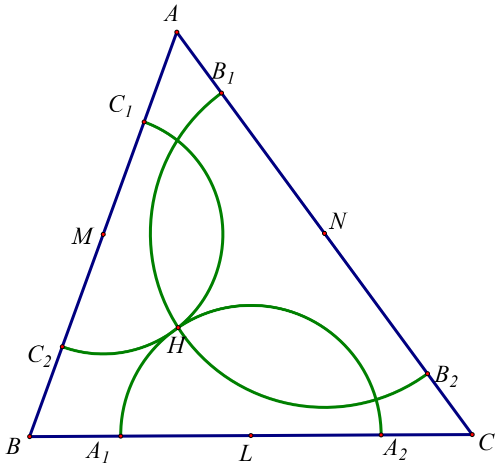
**Example 5.4 (IMO 1995 P1)**

Let $A$, $B$, $C$, $D$ be four distinct points on a line, in that order. The circles with diameters $AC$ and $BD$ intersect at $X$ and $Y$. Let $P$ be a point on the line $XY$ such that $P \notin BC$. The line $CP$ intersects the circle with diameter $AC$ at $C$ and $M$, and the line $BP$ intersects the circle with diameter $BD$ at $B$ and $N$. Prove that the lines $AM$, $DN$, $XY$ are concurrent.

**Exercise**

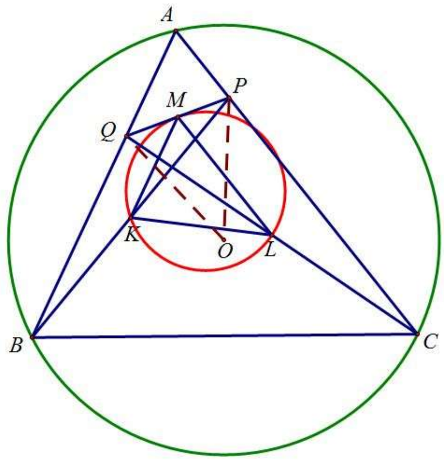
**Problem 1 (IMO 2008 #1)**

An acute-angled triangle $ABC$ has orthocenter $H$. The circle passing through $H$ with centre the midpoint of $BC$ intersects the line $BC$ at $A_1$ and $A_2$. Similarly, the circle passing through $H$ with centre the midpoint of $CA$ intersects the line $CA$ at $B_1$ and $B_2$, and the circle passing through $H$ with centre the midpoint of $AB$ intersect the line $AB$ at $C_1$ and $C_2$. Show that $A_1$, $A_2$, $B_1$, $B_2$, $C_1$, $C_2$ lie on a circle.



**Problem 2 (IMO 2009 #2)**

Let $ABC$ be a triangle with circumcentre $O$. The points $P$ and $Q$ are interior points of the sides $CA$ and $AB$, respectively. Let $K$, $L$ and $M$ be the midpoints of the segments $BP$, $CQ$ and $PQ$, respectively, and let $\Gamma$ be the circle passing through $K$, $L$ and $M$. Suppose that the line $PQ$ is tangent to the circle $\Gamma$. Prove that $OP = OQ$.



**Problem 3 (CGMO 2020 #1)**

As shown in the figure, let $ABCD$ be a quadrilateral with $AB = AD$, $CB = CD$ and $\angle ABC = 90^\circ$. Points $E$, $F$ lie on segments $AB$, $AD$ respectively. Points $P$, Q lie on segment $EF$ ($P$ lies between $E$ and $Q$), such that $\dfrac{AE}{EP} = \dfrac{AF}{FQ}$. Points $X$, $Y$ lie on segments $CP$, $CQ$, respectively, such that $BX \perp CP$, $DY \perp CQ$. Show that $X$, $P$, $Q$, $Y$ are concyclic.